

Statistical Vocabularies in Eurostat Tables

Francesco Pittacolo

1 Introduction

This project concerns the extraction and organization of textual information from statistical tables. The data used in the project are Eurostat tables from the datasets used in the experiments of the STAR paper [1]. Each table contains numerical observations together with textual information that describes them, such as dimension names, dimension values, geographical areas, measures, units, and time information. The title of each table provides additional textual information that is not contained in the table itself.

The main goal of the project is to build a vocabulary from this information and to organize its terms according to their role in the tables. The first part of the work therefore focuses on extracting dates, textual values, and geographical names from the tables and their titles. After removing the geographical and temporal information, the remaining terms form the vocabulary that is used in the following steps.

The vocabulary is then partitioned into four categories: measures M , dimension names N , dimension values A , and unit information U . The identified measures are subsequently organized into broader domains, such as agriculture, transport, or industry.

The last part of the project considers relationships between individual measures. The objective is to identify different types of semantic relationships, for example when two measures refer to the same concept, when one is a more specific version of another, or when two measures describe different aspects of the same phenomenon.

The report describes the implementation of these steps in the order in which they are performed. For each step, it explains the method used, the relevant scripts and resources, and the resulting output. The report also describes the quality evaluation carried out for the vocabulary classification, domain assignment, and semantic relationships.

2 Datasets

Two related Eurostat datasets are used in the project: a full dataset containing 7,605 tables and a subset containing 2,000 tables. Each table is stored as a CSV file containing textual information and numerical observations.

Both datasets are processed by the pipeline, and their results are kept separate. Additional resources used by the different Steps are introduced in the corresponding sections where they are needed.

3 Overview of the Methodology

The project is organized into eight Steps, following the order in which the pipeline is executed:

1. **Temporal extraction:** identify $D(t)$, the time intervals appearing in the top line of each table.
2. **Textual extraction:** extract $S(t)$, the distinct non-numeric strings appearing in the descriptive part of each table.

3. **Geographical identification:** identify $Geo(t)$ within $S(t)$ using the Eurostat NUTS dictionary, and compute $V(t) = S(t) \setminus Geo(t)$.
4. **Title processing:** process the natural-language title of each table, remove temporal and geographical information, and add the remaining terms to $V(t)$.
5. **Global vocabulary:** combine the per-table vocabularies into $V = \bigcup_t V(t)$ while preserving their provenance.
6. **Vocabulary classification:** partition V into measures M , dimension names N , dimension values A , and unit information U , with an additional category O for terms that cannot be assigned reliably.
7. **Measure domains:** assign the measures in M to broader statistical domains using lexical and semantic information, with an LLM used for unclear cases.
8. **Semantic relationships:** identify relationships between pairs of measures using semantic similarity, structural rules, and an LLM for unresolved cases.

The first six Steps focus on extracting and organizing the vocabulary from the structure and textual content of the tables. The last two Steps operate on the resulting measures and add semantic information through domain assignment and pairwise relationship identification.

The implementation corresponding to Steps 1-8 is available in the GitHub repository (<https://github.com/francesco-pittacolo/statistical-vocabularies-eurostat>) under the `src/` directory. The scripts follow the same ordering as the pipeline described in this report, from `src/step1_extract_d_t.py` to `src/step8_semantic_relationships.py`.

The results obtained from the two datasets are presented and compared in Section 13 Results.

4 Step 1: Extraction of Temporal Information $D(t)$

The first Step extracts the temporal information $D(t)$ from the column headers of each table. The script reads the header row of each CSV file and checks each column name against a set of predefined temporal patterns. The headers that match one of these patterns are collected for the corresponding table.

4.1 Detection Rules

A column header is included in $D(t)$ when it matches one of the following patterns:

- **Year:** a four-digit year between 1900 and 2100;
- **Year-month:** a year followed by a two-digit month, such as 2008-01;
- **Date:** a year, month, and day, such as 2020-01-31;
- **Semester:** a year followed by -S1 or -S2, such as 1985-S1;
- **Quarter:** a four-digit year together with a quarter label Q1, Q2, Q3, or Q4. The order is not fixed, so both Q1 2020 and 2020Q1 are recognized;
- **Month names:** an English or French month name, including accented and unaccented French variants, together with a four-digit year.

The first four formats are detected using regular expressions. For quarters, the script checks separately for a valid year and a Q1-Q4 token. Month names are detected using a predefined list of English and French month names, together with a separate check for the year.

For each table, surrounding whitespace is removed from the detected headers and duplicate values are discarded. The remaining values are sorted and stored in a dictionary indexed by table name, with one list of temporal headers for each table. For example:

```
{
"table_1": ["2000", "2001", "2002"],
"table_2": ["2019-Q1", "2019-Q2", "2019-Q3"]
}
```

The resulting $D(t)$ is then used in Step 2 to identify the temporal columns when extracting the textual information from each table.

4.2 Input Handling and Output

The script processes all CSV tables in the input directory and keeps track of whether temporal information was found for each table. At the end of the execution, it reports the number of tables with detected time information, tables without detected time columns, and tables that could not be processed. The extracted $D(t)$ values are then saved in a JSON file.

4.3 Limitations

The extraction relies on a fixed set of regular-expression and keyword rules, so it only covers the temporal formats defined above. If a header uses a different or unexpected format, it may not be recognized and will therefore not be included in $D(t)$. The script also keeps the detected values in their original form and does not try to interpret or normalize them.

5 Step 2: Extraction of Textual Information $S(t)$

The second step extracts $S(t)$, the set of distinct non-numeric strings appearing in the leftmost columns of each table.

5.1 Extraction

For each table, the column headers are compared with the values in $D(t)$. The first temporal header found is used as the boundary: columns before it are considered descriptive and their textual values are collected. Numeric entries are ignored as well, even if they occur in a column that was not identified as temporal.

When no temporal header is detected, the boundary is instead inferred from the structure of the header, using the separator that marks the descriptive part of the Eurostat table. This fallback is used only to identify the relevant columns and does not modify $D(t)$.

The extracted values are cleaned by removing empty cells, missing-value markers, and numeric entries. Duplicates are removed within each column. The result is kept grouped by column, so that the extracted strings remain associated with the dimension from which they originate. For example:

```
{
"Type of vessel": ["Cargo vessel", "Fishing vessel"],
"Unit of measure": ["Million tonnes"]
}
```

This representation is useful for the next steps, where the column names and their values have to be distinguished. The resulting $S(t)$ is saved for all tables and used to construct the vocabulary in the following steps.

6 Step 3: Geographical Terms $Geo(t)$ and $V(t)$

The third step identifies the geographical terms contained in $S(t)$ and separates them from the remaining vocabulary. For this purpose, a reference set of geographical terms is built from the Eurostat NUTS 2021-2024 dataset [2]. It includes country codes, NUTS codes, NUTS labels, country names, and the available Cyrillic and Greek variants. These resources are used to recognize geographical values without relying on the context of a specific table.

6.1 Normalization and Matching

Before comparison, both the reference terms and the values extracted in $S(t)$ are normalized. The normalization converts the text to lowercase and allows parenthesized annotations to be ignored when they are not part of the geographical name.

A value is classified as geographical when its normalized form is found in the reference set. The original value from the table is kept in the output, so that the extracted $Geo(t)$ preserves the spelling used in the source data.

For each table, all matching values are collected in $Geo(t)$. The remaining values are kept in $V(t)$, still grouped by their original dimension. If a dimension contains only geographical values, it is therefore removed from $V(t)$ for that table.

The resulting $Geo(t)$ and $V(t)$ are used in the following step, where the table titles are processed and the remaining vocabulary is completed.

7 Step 4: Processing Titles and Extending $V(t)$

The fourth step processes the natural-language title $\tau(t)$ of each table. Following the same principle used for $S(t)$, geographical and temporal information is separated from the title, and the remaining text is added to $V(t)$. The title remainder is stored under a synthetic pseudo-dimension named **TITLE**, which is kept distinct from real dimension names throughout the rest of the pipeline.

7.1 Removing Geography

Geographical expressions are identified using the same Eurostat NUTS reference used in Step 3. NUTS labels, country names, and available transliterated variants are matched directly in the title text. Longer geographical expressions are processed before shorter ones to avoid partial matches. Country and NUTS codes are not treated as independent geographical expressions in the title.

7.2 Removing Dates

Temporal expressions are removed using an ordered set of patterns covering year ranges such as 2004–2006, as well as expressions such as **from** 2004 **to** 2006, **since** 2004, **until** 2004, **up to** 2004, and **between** 2004 **and** 2006. Individual years, year-month expressions, quarters, semesters, and dates containing a month and day are also removed when they represent temporal information.

Index-base expressions such as (2015 = 100) are protected before the temporal patterns are applied, since the year in this expression is part of the definition of the measure rather than a temporal qualifier. The protected expressions are restored after date removal.

7.3 Cleanup and Addition to $V(t)$

After geographical and temporal information has been removed, a final cleanup pass collapses whitespace, removes empty parenthetical groups, fixes doubled or dangling hyphens, and strips unnecessary leading or trailing punctuation.

If the resulting title remainder is non-empty, it is added to $V(t)$ under the **TITLE** pseudo-dimension. No semantic classification is performed at this Step: the remainder is preserved as vocabulary information and is classified in the subsequent vocabulary-partitioning step.

8 Step 5: Building the Global Vocabulary V

The fifth step builds the total vocabulary V from the vocabularies $V(t)$ obtained for the individual tables. V is defined as the set of distinct terms appearing in one or more tables:

$$V = \bigcup_t V(t).$$

For each term, the implementation also keeps provenance information, namely the list of tables in which the term occurs and the contexts in which it appears. If a term is used as a dimension name, the context "**DIMENSION_NAME**" is recorded. For values associated with a dimension, the name of that dimension is recorded as the context. The synthetic **TITLE** key is not considered a dimension name, but values extracted from titles are associated with the **TITLE** context.

Duplicate terms, table entries, and contexts are removed, and the resulting lists are sorted for reproducibility. The resulting vocabulary V , together with this provenance information, is used in the following steps to classify and organize the extracted terms.

9 Step 6: Classification into M , N , A , U

The sixth step classifies the terms in the global vocabulary V into four categories: measures M , dimension names N , dimension values A , and unit information U .

The classification is based on the structural information available in the per-table vocabularies $V(t)$ and in the provenance information collected in Step 5. No embeddings or LLMs are used at this Step, making the procedure fully deterministic.

9.1 Classification Criteria

Dimension names are obtained from the keys of each $V(t)$, excluding the synthetic **TITLE** key introduced in Step 4. All remaining keys are considered dimension names.

The values associated with each non-title dimension are then examined according to the role of the dimension. Values belonging to unit or frequency-related dimensions are considered candidates for U . The implementation recognizes the dimension names **unit**, **unit of measure**, **measurement unit**, **time frequency**, and **frequency**, using a case-insensitive comparison. Values belonging to all other dimensions are considered dimension-value candidates for A .

Terms associated with the **TITLE** key are collected as measure candidates. These are the title remainders produced in Step 4, after temporal and geographical information have been removed. They provide the main source of candidates for M , since the remaining title text describes the statistical content of the table.

The candidate sets are not necessarily disjoint: the same term may occur in different roles across different tables. A precedence rule is therefore used to assign each term in V to a single category.

9.2 Final Classification

The categories are considered in the following order:

$$N \rightarrow U \rightarrow A \rightarrow M.$$

For each term in V , the classification first checks whether it occurs as a dimension name. If so, it is assigned to N . Otherwise, if it occurs as a value of a unit- or frequency-related dimension, it is assigned to U . If neither condition applies, a term occurring as a value of another dimension is assigned to A . Finally, if it occurs among the title remainders, it is assigned to M .

This precedence resolves overlaps between the candidate sets. Dimension-name evidence is given priority, followed by unit and frequency information, dimension-value evidence, and finally title evidence.

Terms that do not match any of the four candidate sets are assigned to an additional category O . This category serves as a fallback for terms that cannot be assigned to M , N , A , or U .

9.3 Output

The classification is saved in JSON format together with the provenance information associated with each term. The four vocabularies are also exported as separate CSV files, with one term per row under a `term` column. Terms assigned to the fallback category O are exported to a separate CSV file when this category is not empty.

The resulting measure vocabulary M is used in Step 7 to organize the measures into broader statistical domains.

10 Step 7: Grouping Measures into Statistical Domains

The seventh step assigns each measure in M to a statistical domain. Fifteen domains are defined manually: agriculture, transport, industry, energy, environment, population, health, labour, education, economy, science and technology, crime, tourism, construction, and social.

The domains are described through two external JSON resources. The first contains a short textual description for each domain and is used for semantic embedding. The second contains weighted keywords for each domain and is used for dictionary-based matching. Keeping these resources separate from the classification code makes the domain definitions editable without changing the classification procedure.

10.1 Dictionary and Embedding Scores

For each measure and each domain, two independent scores are computed.

The dictionary score is obtained by matching the weighted domain keywords against the measure text using word-boundary regular expressions. The matched weights are summed to obtain a raw score. Since the possible scores differ between domains, the raw score is normalized using a domain-specific calibration value. This value is based on the 95th percentile of the non-zero raw scores observed across the measures when at least 15 measures produce a non-zero score; otherwise, the theoretical maximum score of the domain is used. Scores below a minimum threshold are treated as zero to reduce the effect of isolated keyword matches.

The embedding score is computed from the semantic similarity between the measure text and the textual description of each domain. Both are embedded using `all-MiniLM-L6-v2`, and cosine similarity is used as the similarity measure. The resulting score is restricted to the interval $[0, 1]$.

The two signals are combined using the following weighted score:

$$Score(d, m) = 0.4 Score_{\text{dict}}(d, m) + 0.6 Score_{\text{emb}}(d, m).$$

10.2 Automatic Decision

The classifier does not simply select the domain with the highest blended score. The dictionary and embedding signals are first considered separately.

For each signal, the top two domains are identified. The difference between their scores is used as a margin, which indicates how clearly the first domain stands out from the second. A signal is considered clear when its top score is high enough and the margin over the runner-up is sufficiently large.

When the two signals agree on the same domain and at least one of them is clear, the measure is classified automatically. When they disagree, the classifier uses the domain from the signal that is clear, provided that only one signal meets the criteria. If both signals are clear but point to different domains, or if neither signal is clear, the measure is sent to the LLM instead of making an uncertain automatic decision.

A secondary domain is also recorded when another domain receives a sufficiently high blended score and is close enough to the selected domain.

10.3 LLM Fallback

Measures that cannot be classified confidently by the automatic procedure are processed by a large language model through the Groq API. The model used is `openai/gpt-oss-120b` with temperature set to zero.

The model receives the list of the fifteen allowed domains and must return one domain and a confidence value between 0 and 1. An optional parameter can also be enabled to request a short explanation for the classification. The result is accepted only when the returned domain belongs to the predefined list and the confidence value is valid. This prevents the LLM from introducing new domains or malformed classifications.

Temporary request failures are retried with exponential backoff. If the API daily limit is reached, the LLM phase is stopped and the remaining measures are kept explicitly as unresolved rather than being assigned by guessing.

LLM results are cached locally using a hash of the measure text. Repeated executions can therefore reuse previously obtained classifications without issuing the same request again.

10.4 Output

The procedure produces the domain assignments together with detailed classification information, including the dictionary and embedding scores, the decision reason, the confidence, and the method used. A separate output contains measures that remain unresolved.

11 Step 8: Semantic Relationships Between Measures

The eighth step identifies semantic relationships between pairs of measures in M . Six relation types are considered: `same_concept`, `synonym`, `subtype`, `complementary`, `different_aspects`, and `related`.

Classifying every possible pair would be unnecessarily expensive and would introduce a large number of unrelated pairs. The procedure therefore first constructs a smaller candidate set using semantic and lexical retrieval. The candidate pairs are then analysed using deterministic structural rules and, for sufficiently plausible but unresolved cases, an LLM classifier.

11.1 Candidate Generation

Candidate generation combines two retrieval channels. The first uses semantic embeddings produced with `all-MiniLM-L6-v2`. For every measure, the five nearest neighbours are retrieved

using cosine similarity between normalized embeddings. Pairs with an embedding similarity below 0.55 are discarded from this retrieval channel.

Domain information from Step 7 is used conservatively during embedding retrieval. When two measures have different assigned domains, the pair is retained only when its embedding similarity is at least 0.84. This avoids sending weak cross-domain similarities to the later classification phases while still allowing strongly related measures from different statistical domains.

A second retrieval channel is based on TF-IDF representations of the measure texts. For each measure, the five nearest lexical neighbours are retrieved, with a minimum TF-IDF similarity of 0.18. Unlike the embedding channel, this lexical retrieval is not subject to the hard domain filter. Its purpose is to improve candidate coverage, including pairs whose wording is strongly related even when they are not among the closest embedding neighbours.

Pairs retrieved by either channel are merged into a single candidate set. For each pair, several similarity features are computed: embedding similarity, TF-IDF lexical similarity, token overlap, token containment, and sequence similarity. These features are combined into the semantic score

$$\text{Score} = 0.62 \text{Emb} + 0.16 \text{Lex} + 0.10 \text{Overlap} + 0.07 \text{Containment} + 0.05 \text{Sequence}.$$

Here, Emb is the embedding similarity, Lex is the TF-IDF lexical similarity, Overlap is the token-overlap score, Containment is the token-containment score, and Sequence is the sequence-similarity score.

Pairs retrieved through the embedding channel with a semantic score of at least 0.64 are treated as semantic candidates for further classification. Pairs retrieved only through TF-IDF, or embedding pairs below this threshold, are retained in the candidate output but are not sent directly to the LLM. They may still be accepted when a sufficiently strong deterministic rule applies. Keeping these retrieval-only pairs improves candidate-generation coverage without increasing the number of LLM requests unnecessarily.

11.2 Structural Analysis

Semantic similarity alone is not sufficient to determine the type of relationship. Two measures can have very similar wording while differing in a qualifier that changes their meaning.

The classifier therefore extracts structural information from the original measure texts, including geographic levels, classification revisions, units, numeric restrictions, and other recurring qualifiers. When the same qualifier category occurs with incompatible values, automatic `same_concept` and `subtype` decisions are blocked.

An additional distinction is made for expressions of the form `by X`. In Eurostat measure names, these expressions usually represent a disaggregation dimension rather than a change in the measured indicator. For example, a measure reported `by age` and the corresponding measure reported `by sex` may still represent the same statistical concept. Such cases are therefore treated as evidence for `same_concept`, but are assigned a borderline deterministic confidence rather than being accepted automatically. They are passed to the LLM when the remaining semantic evidence is sufficient.

This also prevents a `by X` suffix from being interpreted automatically as a subtype. A subtype instead requires a genuine restriction of the measured phenomenon, population, category, or scope.

11.3 Learning Recurring Patterns

Before applying the final relationship rules, the candidate set is analysed for recurring token substitutions. Candidate pairs with small lexical differences are compared in order to identify substitutions that occur in multiple contexts.

A recurring substitution can be interpreted as a synonym pattern when the two terms express alternative terminology for the same concept. Alternatively, it can be interpreted as a dimension pattern when it systematically changes one aspect of otherwise similar measures. Patterns are retained only when they satisfy minimum support, term-similarity, context-similarity, and embedding-similarity conditions.

The purpose of this step is to reuse regularities observed in the vocabulary instead of treating every lexical variation independently.

11.4 Deterministic Rule Cascade

Each candidate pair is processed by a fixed sequence of deterministic rules. A rule produces both a relation and a confidence value. The result is accepted automatically only when its confidence reaches the required threshold.

The cascade contains the following main cases:

1. **same_concept**: exact normalized identity and token-reordered forms can be accepted automatically when no structural conflict is present. Measures sharing the same indicator but differing only in a **by X** breakdown are instead treated as borderline **same_concept** candidates and are normally left for LLM verification.
2. **subtype**: assigned when one measure represents a genuinely narrower population, phenomenon, category, or scope. The rule uses token containment together with embedding and structural similarity and rejects cases with incompatible qualifiers. A difference consisting only of a disaggregation dimension is not sufficient to establish a subtype relation.
3. **synonym**: assigned when a sufficiently supported recurring substitution pattern indicates alternative terminology for the same underlying concept.
4. **different_aspects**: assigned when a learned dimension pattern or a strong structural pattern indicates distinct components, quantities, characteristics, categories, or outcomes of the same underlying phenomenon.
5. **related**: a deterministic rule can identify broadly related candidates, but this generic relation is not accepted automatically. It is used only as evidence for subsequent LLM classification.

The automatic confidence thresholds are relation-specific. In particular, **same_concept** and **subtype** require a confidence of at least 0.85, while automatically learned **synonym** and **different_aspects** relations use stricter thresholds. Neither **complementary** nor the generic **related** relation is accepted directly from deterministic rules.

11.5 LLM Classification

The LLM is used only for semantic candidates for which deterministic evidence is insufficient for a safe automatic decision. This avoids classifying every retrieved pair with an LLM while also avoiding an arbitrary fixed limit on the number of pairs that can be reviewed.

Borderline deterministic predictions of **same_concept**, **synonym**, **subtype**, and **different_aspects** are eligible for LLM verification. Otherwise, an embedding candidate must also satisfy minimum semantic and lexical evidence. For same-domain or domain-unknown pairs, the thresholds require an embedding similarity of at least 0.76, a semantic score of at least 0.74, and a lexical or token-overlap signal of at least 0.10. Cross-domain pairs use stricter embedding and semantic thresholds.

Eligible pairs are sent to `openai/gpt-oss-120b` in batches of 40. A shared legend is used inside each batch so that every measure text appears only once, while individual pairs refer to the corresponding identifiers.

The prompt defines the six possible relations explicitly. `same_concept` represents the same measured indicator even when its presentation, frequency, or disaggregation differs. `subtype` requires a genuine restriction of the measured phenomenon, population, category, or scope. `different_aspects` represents distinct components, characteristics, categories, quantities, or outcomes of the same underlying phenomenon. `complementary` is used for distinct statistical indicators whose meanings are naturally connected and which are useful when interpreted together. The generic `related` label is reserved for meaningful semantic relationships that are not adequately described by one of the five more specific relation types. Sharing only a broad statistical topic is not sufficient for assigning a relationship.

The model must return exactly one allowed relation and a confidence value between 0 and 1. Returned pair identifiers, labels, and confidence values are validated before use. LLM classifications are accepted only when their confidence is at least 0.70. Structural safeguards are additionally applied to proposed `same_concept` and `subtype` relations.

LLM results are cached using an identifier derived from the measure pair, the model, and the classifier version. This allows repeated executions of the same configuration to reuse previous classifications while keeping results obtained with different prompt versions separate.

Pairs that do not satisfy a deterministic rule or the LLM acceptance conditions remain unresolved. No generic low-confidence relationship is inserted as a fallback.

11.6 Output

The procedure produces the final semantic relationships together with the complete candidate set, unresolved semantic candidates, summary statistics, a graph representation, and the learned substitution patterns.

For each accepted relationship, the output records the two measures, the assigned relation, confidence, relevant similarity scores, domain information, and the method that produced the classification. Candidate entries also retain their retrieval source, allowing embedding- and TF-IDF-based candidate generation to be analysed separately.

12 Evaluation

The quality evaluation focuses on Steps 6, 7 and 8. The evaluation procedures use manually annotated samples produced separately from the automatic classification process. The corresponding results are presented in Section 13.

12.1 Step 6

The evaluation of the vocabulary classification uses two complementary samples drawn from the vocabulary obtained from the 2,000-table dataset. The first is a fixed-seed random sample of 100 terms drawn directly from V , prior to classification. Each sampled term is provided together with its source tables and contexts, while the `gold_category` field is left empty for manual annotation.

The second sample is stratified by predicted category. It contains 25 terms from each of M , N , A , and U , for a total of 100 terms. The sampled terms are shuffled before annotation so that the predicted category is not visible during the annotation process. This sample provides sufficient examples for the evaluation of each category, including less frequent categories such as U .

For both samples, the evaluation computes overall accuracy, precision, recall, F1-score, macro-F1, and a confusion matrix. The random sample is used as the main estimate of overall performance, while the stratified sample provides a balanced evaluation of the individual categories.

12.2 Step 7

The evaluation of domain classification uses a manually annotated sample of 150 measures drawn from the measure vocabulary obtained from the 2,000-table dataset. The classified measures are ordered by their final blended score and divided into three score regions, with 50 measures sampled from each region using a fixed random seed.

Annotation is performed manually under blind conditions: only the measure text is available during annotation, while the predicted domains, score, classification method, and confidence are kept hidden. These values are stored separately and combined with the reference annotations only during evaluation, reducing the risk that the system predictions influence the reference labels.

The primary domain is evaluated using overall accuracy and macro-F1, together with precision, recall, and F1-score for each domain and a confusion matrix. Accuracy is also analysed by blended-score interval and by classification method, distinguishing the dictionary-and-embedding approach from the LLM-based classifier. This comparison is descriptive, since the two methods operate on different subsets of measures.

Secondary-domain assignment is treated separately as a diagnostic analysis. A secondary domain is manually assigned only when a measure has a substantive relationship with a second statistical domain, rather than when another domain appears only as a disaggregation dimension. The diagnostic records the frequency of reference and predicted secondary domains, exact agreement, and cases in which the reference primary domain is recovered as the predicted secondary domain.

12.3 Step 8

The semantic relationships produced in Step 8 are evaluated on a manually annotated sample of 100 predicted pairs drawn from the relationships identified among measures in the full 7,605-table dataset.

The sampling frame contains predictions belonging to the five specific relation types: **same_concept**, **synonym**, **subtype**, **complementary**, and **different_aspects**. The sample is constructed approximately evenly across the predicted relation types, with a target of 20 pairs per relation when enough examples are available.

Annotation is performed under blind conditions. Only the two measure texts are available during annotation, while the predicted relation, classification method, and confidence are kept hidden and combined with the reference labels only during scoring. The reference label may also be **related** or **unrelated** when none of the five specific relations is appropriate.

Relationship classification is evaluated using the proportion of semantically valid pairs, exact relation accuracy, precision, recall, and F1-score for each of the five specific relation types. Macro-F1 is computed over these five relations. Results are also reported by classification method as a descriptive comparison between deterministic and LLM-based decisions.

As an additional diagnostic, deterministic classifications are also reviewed using an LLM-based judge on a small sample drawn from each deterministic classification method. This diagnostic is not treated as reference ground truth.

12.4 Limitations

The evaluation relies on manually annotated samples rather than exhaustive ground truth for the complete vocabulary and all measure pairs. Consequently, the reported scores provide evidence about the quality of the implemented methods but do not establish their performance on every possible input. Additional validation by an independent annotator and an inter-annotator agreement measure such as Cohen’s κ would further strengthen the evaluation, especially for the more subjective classifications in Steps 7 and 8.

13 Results

13.1 Step 1: Temporal Information

Step 1 was executed on both datasets. In the full dataset of 7,605 tables, temporal information $D(t)$ was detected in 7,602 cases, while 3 inputs contained no recognized temporal columns.

For the subset of 2,000 tables, temporal information was detected in 1,999 tables, with only 1 table containing no recognized temporal column.

Table 1: Step 1 temporal-extraction results.

Dataset	Processed	With $D(t)$	Without $D(t)$	Errors
7,605-table dataset	7,605	7,602	3	0
2,000-table dataset	2,000	1,999	1	0

Temporal information was therefore identified in approximately 99.96% of the inputs in the full dataset and 99.95% of the tables in the 2,000-table subset. The very similar coverage observed in the two datasets indicates that the temporal extraction rules behave consistently across the two datasets.

In the 7,605-table dataset, the inputs without detected temporal columns were `aact_ali01.csv`, `env_wat_ltaa.csv`, and `ten00001.csv`. In the 2,000-table dataset, only `ten00001.csv` had no detected temporal column.

Overall, temporal information was detected for almost all inputs in both datasets.

13.2 Step 2: Textual Information

Step 2 was executed on both datasets to construct $S(t)$ from the non-numeric values appearing in the descriptive part of each table.

For the full dataset, 7,605 input files were examined and a non-empty $S(t)$ was obtained for 7,604 of them. Two valid tables, `env_wat_ltaa.csv` and `ten00001.csv`, had no temporal information detected in Step 1. In both cases, the structural fallback used by Step 2 still allowed the descriptive part of the table to be identified and processed.

One input, `aact_ali01.csv`, could not be processed as a regular CSV file, accounting for the single processing error in the full dataset.

For the 2,000-table subset, all 2,000 tables produced a non-empty $S(t)$. Only `ten00001.csv` had no temporal information detected in Step 1, and no processing errors occurred.

Table 2: Step 2 textual-extraction results.

Dataset	Processed	With $S(t)$	Without $D(t)$	Errors
7,605-table dataset	7,605	7,604	2	1
2,000-table dataset	2,000	2,000	1	0

Step 2 therefore produced textual information for all regular tables in the 2,000-table subset and for all processable tables in the full dataset.

The results also show that the absence of a temporal match in Step 1 does not necessarily prevent the extraction of $S(t)$. Temporal detection and textual-boundary identification are therefore kept separate: $D(t)$ remains the primary source for locating the temporal boundary, while the structural fallback is used only when an explicit temporal match is unavailable.

13.3 Step 3: Geographical Terms

Step 3 was executed on both datasets using the same geographical reference vocabulary, containing 3,826 terms derived from the Eurostat NUTS resource together with country names and the available Cyrillic and Greek variants.

In the 2,000-table subset, geographical values were identified in 1,605 tables, corresponding to approximately 80.25% of the dataset. Across these tables, 47,236 geographical values were identified, with 486 distinct values. After removing geographical information, all 2,000 tables retained a non-empty $V(t)$. The resulting per-table vocabularies contained 37,010 vocabulary values across tables, corresponding to 7,285 distinct values.

In the full dataset, geographical values were found in 7,011 of the 7,605 table entries, corresponding to approximately 92.19%. The procedure identified 374,478 geographical values across tables and 3,609 distinct geographical values. A non-empty $V(t)$ was obtained for 7,604 table entries. The resulting vocabularies contained 590,150 vocabulary values across tables, with 127,179 distinct values.

Table 3: Step 3 geographical-processing results.

Result	7,605-table dataset	2,000-table subset
Tables with $Geo(t)$	7,011	1,605
Geographic values across tables	374,478	47,236
Distinct geographic values	3,609	486
Tables with non-empty $V(t)$	7,604	2,000
Vocabulary values across tables	590,150	37,010
Distinct vocabulary values	127,179	7,285

The results show that geographical information occurs in a large proportion of the Eurostat tables, particularly in the full dataset. The larger dataset also contains substantially more distinct geographical values. After geographical terms are removed, however, almost every table still retains descriptive vocabulary in $V(t)$, which is then extended with title information in Step 4.

13.4 Step 4: Title Information

Step 4 was executed on both datasets by processing the title associated with each table, removing geographical and temporal information, and adding the remaining text to $V(t)$ under the **TITLE** pseudo-dimension.

For the 2,000-table subset, all 2,000 titles were found and all produced a non-empty remainder that was added to $V(t)$. No title was missing and no title became empty after the removal procedure. The resulting title information contained 1,838 distinct remainders.

The same behaviour was observed on the full dataset. All 7,605 titles were found and successfully added to $V(t)$, with no missing titles and no empty remainders. In total, 6,423 distinct title remainders were obtained.

For title processing, the geographical matching vocabulary was restricted to 2,325 terms in order to avoid ambiguous matches caused by geographical codes occurring as ordinary text.

Table 4: Step 4 title-processing results.

Result	7,605-table dataset	2,000-table subset
Titles processed	7,605	2,000
Non-empty title remainders	7,605	2,000
Distinct title remainders	6,423	1,838
Missing titles	0	0

Overall, Step 4 was able to associate and process a title for every table in both datasets. The absence of empty remainders indicates that, after removing temporal and geographical information, each title still retained descriptive content that could contribute to the construction of the vocabulary.

13.5 Step 5: Global Vocabulary

Step 5 merged the per-table vocabularies into a single global vocabulary V , while retaining the tables and contexts in which each term occurs.

For the 2,000-table subset, the resulting vocabulary contained 9,445 distinct terms. Among these, 350 were associated with a dimension-name context, 1,838 occurred in the **TITLE** context, and 3,831 appeared in more than one table.

For the full dataset, the global vocabulary contained 134,157 distinct terms. Of these, 638 were associated with a dimension-name context, 6,423 occurred in the **TITLE** context, and 46,988 appeared in multiple tables.

Table 5: Step 5 global-vocabulary results.

Result	7,605-table dataset	2,000-table dataset
Vocabulary size	134,157	9,445
Dimension-name context	638	350
Terms in TITLE	6,423	1,838
Terms in multiple tables	46,988	3,831

The 7,605-table dataset produces a substantially larger global vocabulary, as expected from its broader set of tables and descriptive values. At the same time, many terms occur in multiple tables, showing that a considerable part of the vocabulary is shared across different Eurostat tables rather than being specific to a single table.

13.6 Step 6: Vocabulary Classification

Step 6 classified every term in the global vocabulary V into one of the categories M , N , A , U , or the fallback category O .

For the full dataset, the vocabulary contained 134,157 terms. Among these, 6,348 were classified as measures M , 641 as dimension names N , 126,453 as dimension values A , and 715 as units U . No term was assigned to the fallback category O .

For the 2,000-table subset, the vocabulary contained 9,445 terms. The classification produced 1,812 measures, 350 dimension names, 7,015 dimension values, and 268 units. Again, no term remained in the fallback category.

Table 6: Step 6 vocabulary-classification results.

Category	7,605-table dataset	2,000-table subset
Measures M	6,348	1,812
Dimension names N	641	350
Dimension values A	126,453	7,015
Units U	715	268
Other O	0	0
Total	134,157	9,445

The classification is dominated by dimension values in both datasets, while measures, dimension names, and units form substantially smaller parts of the global vocabulary. The absence of terms in O indicates that every vocabulary entry matched at least one of the structural categories defined by the classification procedure.

13.6.1 Step 6 Evaluation

The Step 6 classification was evaluated manually using two samples of 100 terms each. The first was a fixed-seed random sample from the global vocabulary, while the second was stratified across the four predicted categories, with 25 terms sampled from each of M , N , A , and U .

In the random sample, the manual reference labels consisted of 23 measures, 4 dimension names, 70 dimension values, and 3 units. All 100 terms were classified correctly, resulting in an accuracy of 100% and a macro F1-score of 100%. Precision, recall, and F1-score were also 100% for each category.

The stratified sample produced the same result. All 25 terms in each of the four categories matched the manual reference labels, yielding an accuracy of 100% and a macro F1-score of 100%.

Table 7: Step 6 manual-evaluation results.

Metric	Random sample	Stratified sample
Sample size	100	100
Accuracy	100%	100%
Macro F1-score	100%	100%

The random evaluation assesses the performance of the classifier under the natural class distribution, although the sample is dominated by dimension values. The stratified evaluation complements this result by giving equal representation to all four classes. No disagreement between the automatic classification and the manual reference labels was observed in either evaluation sample.

13.7 Step 7: Measure Domains

Step 7 assigned the measures identified in Step 6 to the predefined set of 15 statistical domains.

For the 2,000-table subset, all 1,812 measures received a primary domain. The dictionary-and-embedding component directly classified 802 measures, corresponding to 44.26% of the total. The remaining 1,010 measures, or 55.74%, were handled by the LLM-based fallback.

The same procedure classified all 6,348 measures in the full dataset. In this case, 3,126 measures (49.24%) were assigned by the dictionary-and-embedding component and 3,222 (50.76%) by the LLM-based fallback. No unresolved measures remained in either dataset.

Table 8: Step 7 domain-assignment coverage.

Result	7,605-table dataset	2,000-table subset
Measures classified	6,348	1,812
Dictionary and embedding	3,126	802
LLM	3,222	1,010
Unresolved	0	0

The distribution of measures across domains is shown in Table 9. In the full dataset, the largest domains were labour with 1,103 measures, economy with 1,007, and population with 734. In the 2,000-table subset, economy was the largest domain with 351 measures, followed by labour with 229 and transport with 213.

Table 9: Distribution of measures across the Step 7 domains.

Domain	7,605-table dataset	2,000-table subset
Agriculture	412	105
Construction	132	63
Crime	68	74
Economy	1,007	351
Education	410	77
Energy	172	47
Environment	215	97
Health	514	86
Industry	220	54
Labour	1,103	229
Population	734	144
Science and technology	348	65
Social	410	181
Tourism	125	26
Transport	478	213
Total	6,348	1,812

The domain distribution is not uniform in either dataset. The relative importance of some domains also differs between the two datasets.

All LLM classifications required by these runs were already present in the existing cache. Therefore, no new LLM requests were necessary during the reported executions. This affects only execution time and does not change the final domain assignments.

13.7.1 Step 7 Evaluation

The quality of the domain assignment was evaluated manually on a sample of 150 measures. The reference primary domain was assigned without using the predicted domain. A secondary reference domain was also recorded when a measure had a meaningful association with more than one statistical area.

Of the 150 evaluated measures, 118 received the correct primary domain and 32 were classified incorrectly. This corresponds to an accuracy of 78.7% and a macro F1-score of 69.1%.

Table 10: Step 7 primary-domain evaluation.

Metric	Result
Sample size	150
Correct classifications	118
Incorrect classifications	32
Accuracy	78.7%
Macro F1-score	69.1%

Performance varied considerably across domains. Agriculture, crime, economy, science and technology, and transport obtained high F1-scores, whereas education, population, construction, and industry were more difficult for the classifier.

Table 11: Step 7 evaluation by primary domain.

Domain	Support	Precision	Recall	F1-score
Agriculture	8	88.9%	100.0%	94.1%
Construction	2	22.2%	100.0%	36.4%
Crime	3	100.0%	100.0%	100.0%
Economy	28	96.2%	89.3%	92.6%
Education	8	100.0%	12.5%	22.2%
Energy	4	66.7%	100.0%	80.0%
Environment	10	80.0%	80.0%	80.0%
Health	15	91.7%	73.3%	81.5%
Industry	1	0.0%	0.0%	0.0%
Labour	16	70.0%	87.5%	77.8%
Population	7	33.3%	57.1%	42.1%
Science and technology	4	80.0%	100.0%	88.9%
Social	26	94.4%	65.4%	77.3%
Transport	18	94.4%	94.4%	94.4%

The low recall for education indicates that several measures whose main subject was educational were assigned to other domains. Population also showed relatively low precision, indicating that the presence of population terminology can sometimes lead to this domain being selected even when population is only a denominator or a descriptive characteristic. The results for construction and industry should be interpreted cautiously because their manual reference supports were only 2 and 1, respectively.

The two classification methods also showed different performance on the evaluation sample. The LLM classified 83 sampled measures and achieved an accuracy of 85.5%, while the dictionary-and-embedding method classified 67 measures with an accuracy of 70.2%.

Table 12: Step 7 evaluation by classification method.

Method	Evaluated measures	Accuracy
LLM	83	85.5%
Dictionary and embedding	67	70.2%

The internal classification score was also examined by score interval. The relationship between score and accuracy was not monotonic: measures below 0.50 achieved an accuracy of 81.3%, while those between 0.50 and 0.60 achieved 63.6%, and those between 0.60 and 0.70

achieved 69.2%. All 8 measures with scores between 0.70 and 0.80 were classified correctly. No evaluated measure had a score of at least 0.80. The score should therefore be interpreted as an internal decision signal rather than as a calibrated probability of correctness.

Secondary domains were evaluated separately as a diagnostic measure. A secondary reference domain was assigned to 25 of the 150 measures, corresponding to 16.7% of the evaluation sample. The classifier predicted a secondary domain for 11 measures, or 7.3%, and 4 of these secondary predictions exactly matched the manual secondary domain.

In addition, in 4 of the 11 cases containing a predicted secondary domain, the manually assigned primary domain appeared as the predicted secondary domain. This indicates that some primary-domain errors still captured the appropriate statistical area as a secondary association.

Overall, the evaluation shows that the hybrid procedure assigns the correct primary domain to most measures, while also revealing specific difficulties for domains with overlapping terminology or limited representation in the evaluation sample. The secondary-domain analysis is treated as diagnostic rather than as part of the main accuracy measure.

13.8 Step 8: Semantic Relationships

Step 8 was applied to the measures obtained from both datasets. The 2,000-table subset contained 1,812 measures, while the full 7,605-table dataset contained 6,348.

Table 13 summarizes the main stages of the relationship-generation procedure. The full dataset produced substantially more candidate pairs and semantic candidates, as expected from its larger measure vocabulary. In both datasets, only a subset of the retrieved pairs satisfied the semantic-candidate criteria, while the remaining pairs were retained as retrieval-only candidates.

Table 13: Step 8 relationship-generation results.

Result	7,605-table dataset	2,000-table subset
Measures	6,348	1,812
Candidate pairs	27,335	6,547
Semantic candidates	12,367	1,979
Retrieval-only candidates	14,968	4,568
Resolved by deterministic rules	904	146
Sent to the LLM	7,447	1,087
Accepted LLM results	7,397	1,082
Final relationships	8,301	1,228
Unresolved pairs	4,078	753

The structural-analysis phase identified 474 recurring substitutions in the full dataset and 149 in the 2,000-table subset. No learned synonym variant satisfied the acceptance criteria in either dataset, while 36 recurring dimension patterns were retained in the full dataset and 6 in the subset.

The distribution of the final semantic relations differed between the two datasets. In the full dataset, **same_concept** was the most frequent relation, whereas **different_aspects** was the most frequent relation in the 2,000-table subset. In both cases, these two classes together accounted for most of the final relationship set, followed by **subtype**. The remaining relation types were comparatively rare.

Table 14: Distribution of the final Step 8 semantic relationships.

Relation	7,605-table dataset		2,000-table subset	
	Count	Share	Count	Share
same_concept	4,259	51.31%	459	37.38%
different_aspects	2,936	35.37%	537	43.73%
subtype	935	11.26%	210	17.10%
related	114	1.37%	16	1.30%
complementary	53	0.64%	5	0.41%
synonym	4	0.05%	1	0.08%
Total	8,301	100.00%	1,228	100.00%

Most accepted relationships in both datasets were produced through LLM classification, while the deterministic rules accounted for a smaller fraction. In the full dataset, the 904 deterministic relationships consisted mainly of automatic subtype decisions, learned dimension patterns, and token-reordering matches. The same pattern was observed in the 2,000-table subset, where the corresponding deterministic total was 146 relationships.

13.8.1 Step 8 Evaluation

The semantic relationship classification was evaluated on a manually annotated sample of 100 predicted pairs. All 100 sampled pairs were judged to express a valid semantic relationship. Of the reference labels, 99 belonged to one of the five specific relation types, while one pair was assigned the generic **related** relation. No pair was annotated as **unrelated**.

The predicted relation exactly matched the reference relation for 72 of the 100 pairs, corresponding to an exact relation accuracy of 72.0%. The macro F1-score over the five specific relation types was 74.4%.

Table 15: Step 8 relationship-classification evaluation.

Relation	Gold	Predicted	Precision	Recall	F1-score
complementary	21	20	85.0%	81.0%	82.9%
different_aspects	25	25	68.0%	68.0%	68.0%
same_concept	34	30	80.0%	70.6%	75.0%
subtype	14	21	47.6%	71.4%	57.1%
synonym	5	4	100.0%	80.0%	88.9%

Performance varied across relation types. The highest F1-score was obtained for **synonym**, at 88.9%, followed by **complementary** at 82.9%. **same_concept** obtained an F1-score of 75.0%, while **different_aspects** obtained 68.0%. The most difficult relation was **subtype**, with an F1-score of 57.1%. Its recall was 71.4%, but its precision was 47.6%, indicating that **subtype** was over-predicted in the evaluation sample.

Most evaluated pairs were classified by the LLM-based procedure. The LLM accounted for 84 of the 100 sampled relationships and achieved an exact relation accuracy of 76.2%. The automatic subtype rule accounted for 12 pairs and achieved 50.0% exact accuracy. The learned-dimension rule accounted for 3 pairs and achieved 66.7% exact accuracy, while the automatic token-reordering rule occurred in only one sampled pair and did not match the reference relation in that case.

Table 16: Step 8 evaluation by classification method.

Method	Sample size	Exact accuracy
LLM classification	84	76.2%
Automatic subtype	12	50.0%
Automatic learned dimension	3	66.7%
Automatic token reorder	1	0.0%

The comparison between classification methods is descriptive, since the deterministic methods are evaluated on much smaller samples and operate on different subsets of relationships.

As an additional diagnostic, the deterministic Step 8 classifications were checked using an LLM-based judge. Up to 10 relationships were sampled from each deterministic classification method, producing a total sample of 43 pairs. For each pair, the judge received the two measure texts together with the relation assigned by the system and was asked whether it agreed with that classification.

The judge evaluated all 43 sampled pairs without errors and agreed with the assigned relation in 31 cases, corresponding to an agreement rate of 72.1%. The remaining 12 cases received a disagreement. The reported agreement rate refers to this specific execution, since the LLM judge may produce slightly different results across repeated runs even with temperature set to zero. This result is treated only as a diagnostic consistency check and not as independent ground truth, since the judge is itself an LLM and is shown the system prediction. The manually annotated sample therefore remains the primary evaluation of Step 8.

Overall, the evaluation indicates that the system identifies the specific semantic relation correctly for most sampled relationships, while **subtype** remains the main source of classification error.

14 Conclusions

This project developed a complete pipeline for extracting and organizing statistical vocabulary from Eurostat tables. The approach combines structural information from the tables with external geographical resources, lexical and semantic similarity, deterministic rules, and LLM-based classification. The result is a vocabulary in which terms are separated according to their role, measures are grouped into statistical domains, and semantic relationships between measures are identified.

The evaluation shows that the deterministic vocabulary classification performed reliably on the annotated samples, while the domain and relationship classification tasks are more challenging because they require finer semantic distinctions. In particular, overlapping domains and closely related relation types remain the main sources of ambiguity. The evaluation is based on manually annotated samples rather than exhaustive ground truth, so larger and independently annotated samples would provide a stronger assessment of the system. Overall, the pipeline provides a reproducible way to transform the textual content of Eurostat tables into a structured statistical vocabulary that can support further semantic analysis.

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